Exercise 24.1

Give an intuitive explanation why the estimator σ_N is biased, and how the bias depends on the number of observations.

Solution:

Sample mean is where the sum squared deviation of the data points are minimised. However, the sample mean is unlikely to exactly equal tht true mean. Any other value that is not the sample mean will have a larger value of the sum-squared deviation, so the expected variance from the true mean must be greater than the variance of the sample mean.

Exercise 24.2

Marginalize over σ and obtain the posterior marginal distribution of the mean μ given the data and a prior for μ and σ.

Solution:

$$
\begin{aligned}
P(\mu|D) &= \frac{P(D|\mu)P(\mu)}{P(D)} \\
P(\mu|D) &\propto P(D|\mu) \\
&= \int_0^\infty P(D|\mu,\sigma)P(\sigma)d\sigma \\
&= \int_0^\infty \prod_{n=1}^{N} \frac{1}{\sqrt{2\pi}\sigma} \exp(-\frac{(x_n-\mu)^2}{2\sigma^2})\frac{1}{\sigma}d\sigma \\
&\propto \int_0^\infty \frac{1}{\sigma^{N+1}} \exp(-\sum_{n=1}^{N} \frac{(x_n-\mu)^2}{2\sigma^2})d\sigma \\
&= \int_0^\infty \frac{1}{\sigma^{N+1}} \exp(-\frac{S_\mu}{2\sigma^2})d\sigma \quad (S_\mu = \sum_{n=1}^{N}(x_n-\mu)^2) \quad \dots(1)
\end{aligned}
$$

Now, if we let $t = \frac{S_\mu}{2\sigma^2}$, $\sigma = \sqrt{\frac{S_\mu}{2t}}$, then

$$
\frac{d\sigma}{dt} = \sqrt{\frac{S_\mu}{2}} \times -\frac{1}{2}t^{-\frac{3}{2}}
$$

$$
\begin{aligned}
(1) &= \int_{\infty}^{0} \left(\frac{S_\mu}{2t}\right)^{-\frac{N}{2}-\frac{1}{2}} \exp(-t) \times -\frac{1}{2} t^{-\frac{3}{2}} \left(\frac{S_\mu}{2}\right)^{\frac{1}{2}} dt \\
&= -\frac{1}{2} \int_{\infty}^{0} \left(\frac{S_\mu}{2t}\right)^{-\frac{N}{2}} \left(\frac{1}{t}\right)^{-\frac{N}{2}+1} \exp(-t) dt \\
&= \frac{1}{2} \left(\frac{S_\mu}{2t}\right)^{-\frac{N}{2}} \int_{0}^{\infty} t^{\frac{N}{2}-1} \exp(-t) dt \\
&= \frac{1}{2} \left(\frac{S_\mu}{2t}\right)^{-\frac{N}{2}} \Gamma(\frac{N}{2})
\end{aligned}
$$

Only considering terms that depend on μ,

$$
P(\mu|D) \propto S_\mu^{-\frac{N}{2}} = \frac{1}{[N(\mu - \bar{x}) + S]^{\frac{N}{2}}}
$$

Exercise 24.3

[This exercise requires macho integration capabilities.] Give the posterior predictive distribution for a new observation x_{N+1} given data $\{x_n\}$ and a prior on (μ, σ), and show how the result reduces to the Student-t distribution in the appropriate limit.

Solution: